

# **Computer Networks**

## *[Computer Network Types]*

# Computer Network Types

- A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.
- A computer network can be categorized by their size.
- A computer network is mainly of four types:
  - LAN(Local Area Network)
  - PAN(Personal Area Network)
  - MAN(Metropolitan Area Network)
  - WAN(Wide Area Network)

# LAN (Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in LAN.
- Local Area Network provides higher security.

# LAN (Local Area Network)



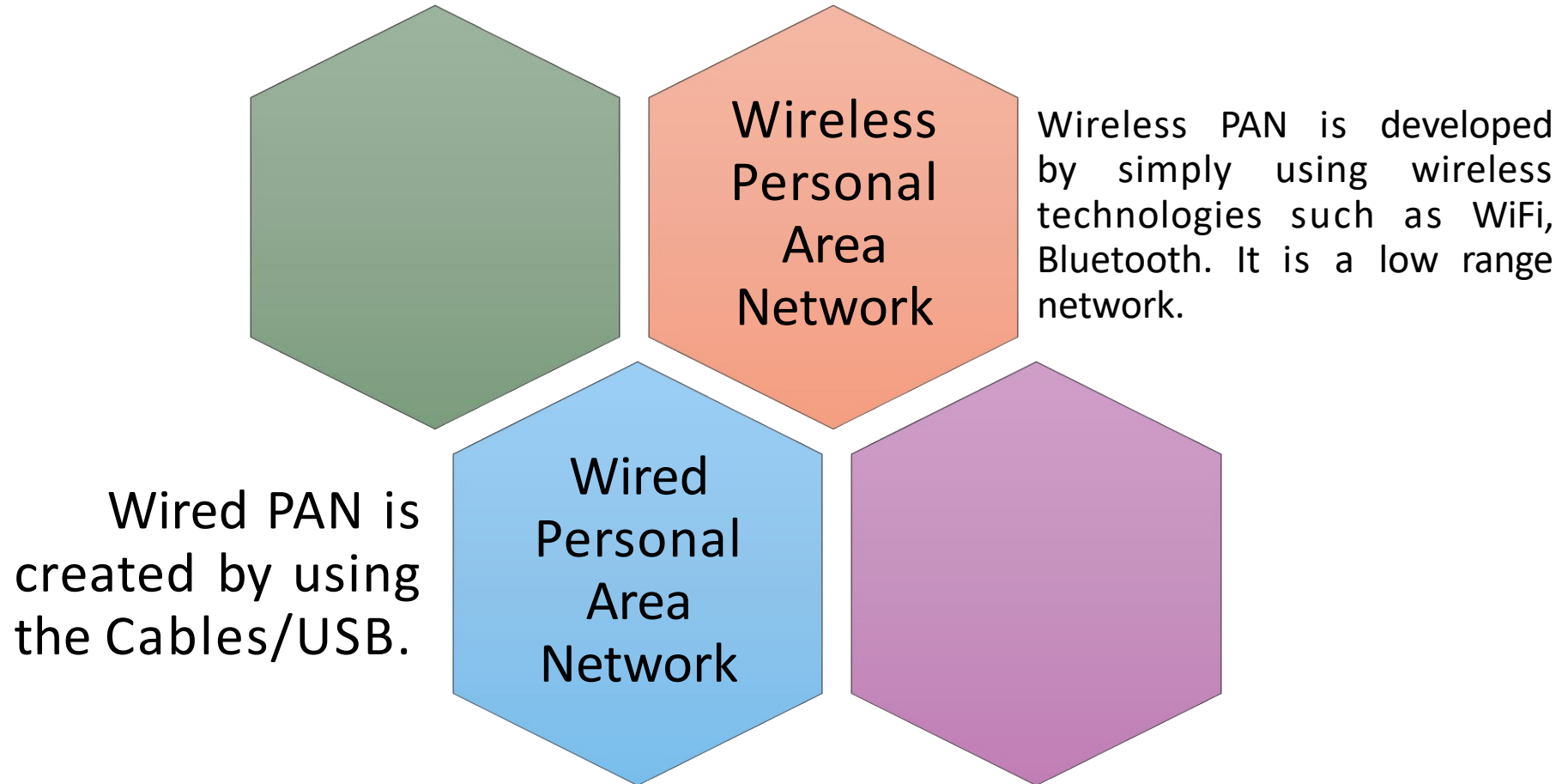
# PAN (Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- PAN is used for connecting the computer devices of personal use.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.

# PAN (Personal Area Network)



# Types of Personal Area Network



# Examples of Personal Area Network

## Body Area Network

- Body Area Network is a network that moves with a person.
- For example, a mobile network moves with a person. Suppose a person establishes a network connection and then creates a connection with another device to share the information.

## Offline Network

- An offline network can be created inside the home, so it is also known as a **home network**.
- A home network is designed to integrate the devices such as printers, computer, television but they are not connected to the internet.

## Small Home Office

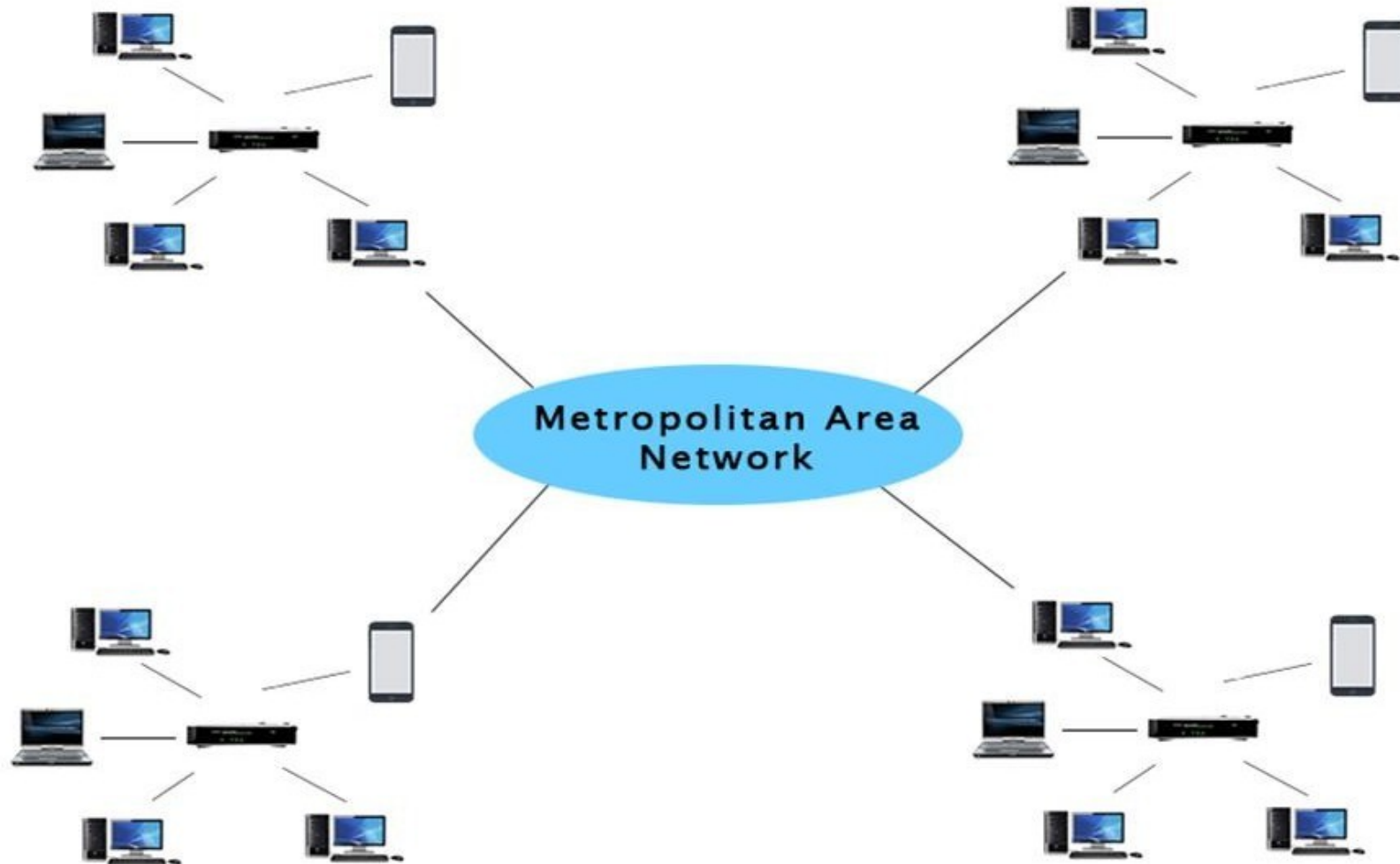
- It is used to connect a variety of devices to the internet and to a corporate network using a VPN.



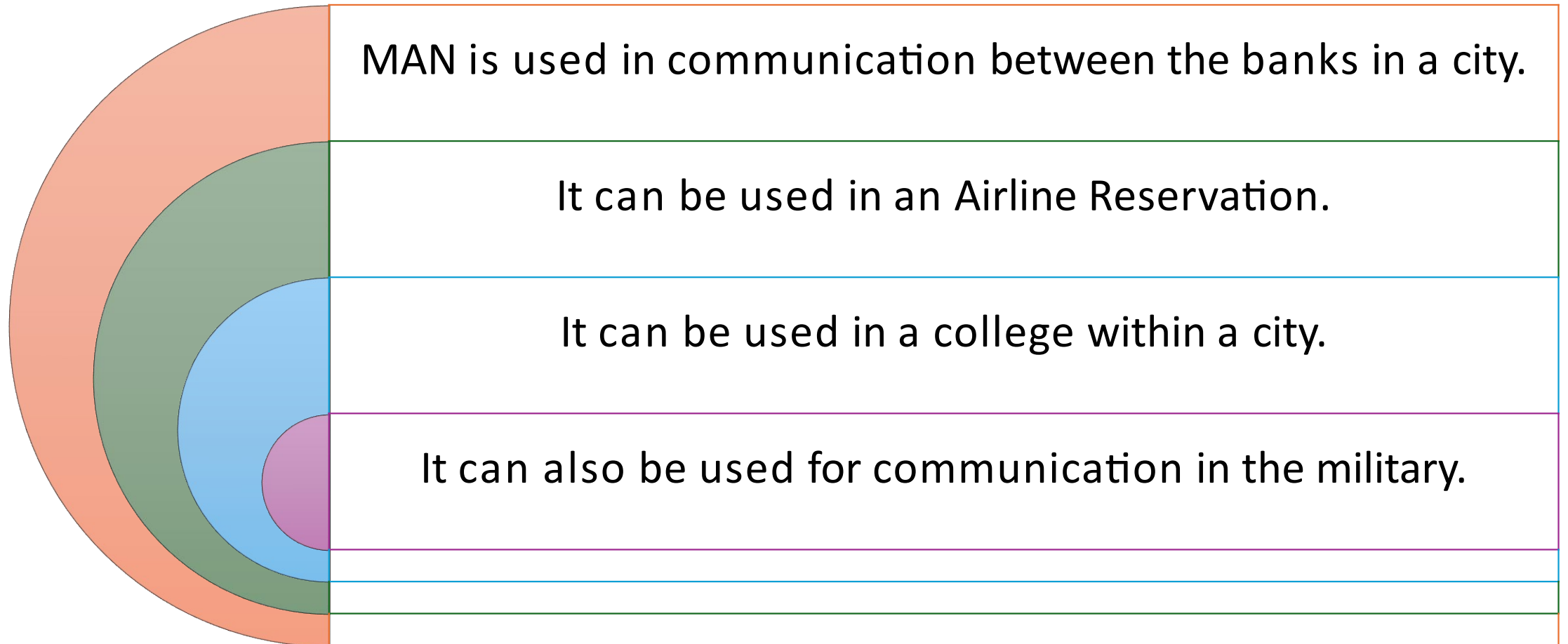
# MAN (Metropolitan Area Network)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has a higher range than Local Area Network (LAN).

# MAN (Metropolitan Area Network)



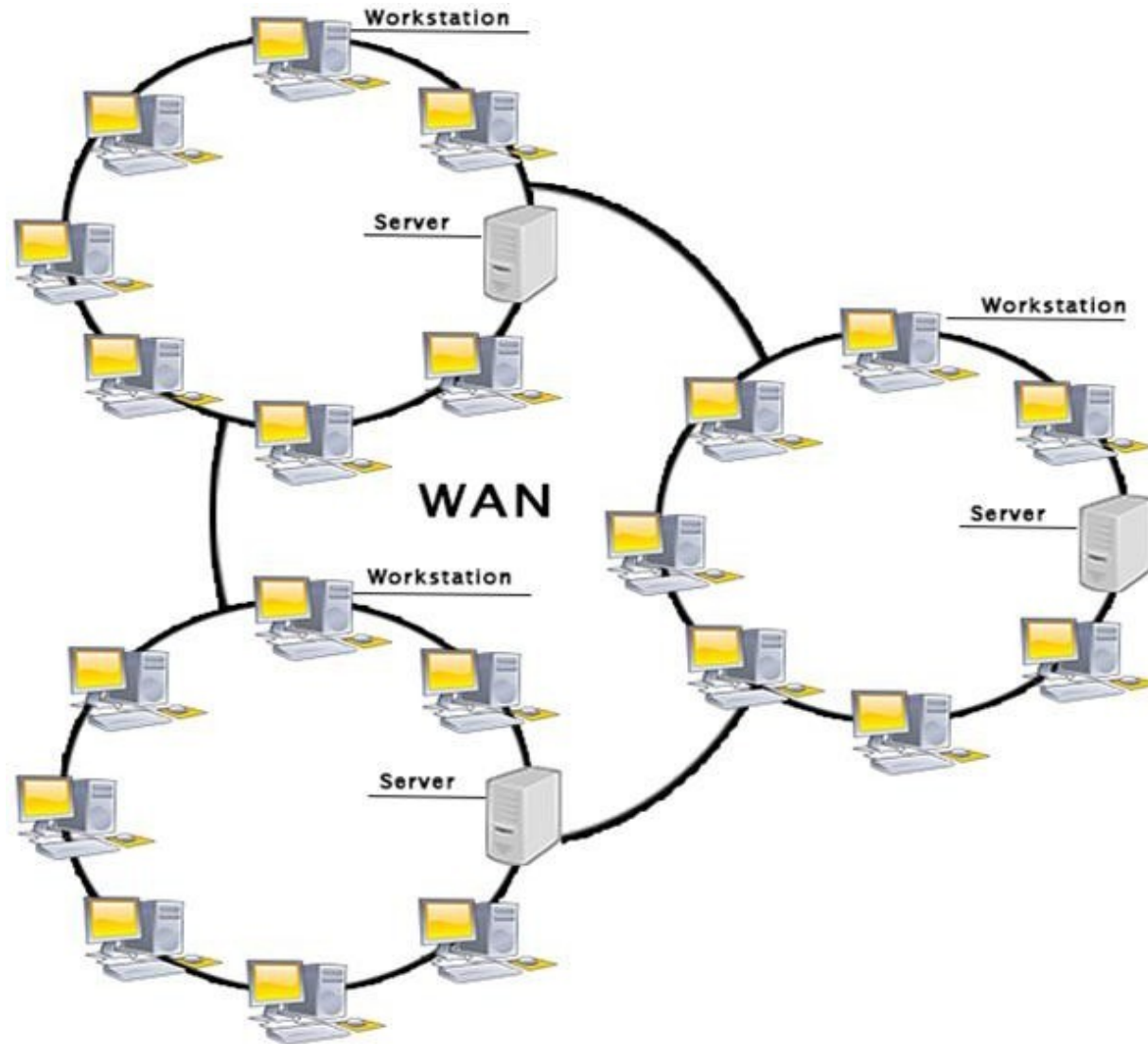
# Uses of Metropolitan Area Network



# WAN (Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A WAN is quite bigger network than the LAN.
- A WAN is not limited to a single location, but it spans over a large geographical area through a telephone line, fiber optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A WAN is widely used in the field of Business, government, and education.

# WAN (Wide Area Network)



# Examples of Wide Area Network

## Mobile Broadband:

- A 4G network is widely used across a region or country.

## Last mile:

- A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.

## Private network:

- A bank provides a private network that connects the 44 offices.
- This network is made by using the telephone leased line provided by the telecom company.

# Advantages of Wide Area Network

|                                    |                                                                                                                                                                                                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geographical area:                 | <ul style="list-style-type: none"><li>• A Wide Area Network provides a large geographical area. Suppose if the branch of our office is in a different city, then we can connect with them through WAN. The internet provides a leased line through which we can connect with another branch.</li></ul> |
| Centralized data:                  | <ul style="list-style-type: none"><li>• In case of WAN network, data is centralized. Therefore, we do not need to buy the emails, files or back up servers.</li></ul>                                                                                                                                  |
| Get updated files:                 | <ul style="list-style-type: none"><li>• Software companies work on the live server. Therefore, the programmers get the updated files within seconds.</li></ul>                                                                                                                                         |
| Exchange messages:                 | <ul style="list-style-type: none"><li>• In a WAN network, messages are transmitted fast. The web application like Facebook, Whatsapp, Skype allows you to communicate with friends.</li></ul>                                                                                                          |
| Sharing of software and resources: | <ul style="list-style-type: none"><li>• In WAN network, we can share the software and other resources like a hard drive, RAM.</li></ul>                                                                                                                                                                |
| Global business:                   | <ul style="list-style-type: none"><li>• We can do the business over the internet globally.</li></ul>                                                                                                                                                                                                   |
| High bandwidth:                    | <ul style="list-style-type: none"><li>• If we use the leased lines for our company then this gives the high bandwidth. The high bandwidth increases the data transfer rate which in turn increases the productivity of our company.</li></ul>                                                          |

# Disadvantages of Wide Area

## Networks

### Security issue:

- A WAN network has more security issues as compared to LAN and MAN network as all the technologies are combined that creates the security problem.

### Needs Firewall s antivirus software:

- The data is transferred on the internet which can be changed or hacked by the hackers, so the firewall needs to be used.
- Some people can inject the virus in our system, so antivirus is needed to protect from such a virus.

### High Setup cost:

- An installation cost of the WAN network is high as it involves the purchasing of routers, switches.

### Troubleshooting problems:

- It covers a large area so fixing the problem is difficult.



# Internetwork

- An internetwork is defined as two or more computer network LANs or WAN or computer network segments are connected using devices, and they are configured by a local addressing scheme. This process is known as **internetworking**.
- An interconnection between public, private, commercial, industrial, or government computer networks can also be defined as **internetworking**.
- An internetworking uses the **internet protocol**.
- The reference model used for internetworking is **Open System Interconnection (OSI)**.

# Types of Internetwork

## Extranet:

- An extranet is a communication network based on the internet protocol such as **Transmission Control protocol and internet protocol**. It is used for information sharing. The access to the extranet is restricted to only those users who have login credentials.
- An extranet is the lowest level of internetworking. It can be categorized as **MAN**, **WAN** or other computer networks. An extranet cannot have a single **LAN**, at least it must have one connection to the external network.

## Intranet:

- An intranet is a private network based on the internet protocol such as **Transmission Control protocol and internet protocol**. An intranet belongs to an organization which is only accessible by the **organization's employee** or members.
- The main aim of the intranet is to share the information and resources among the organization employees. An intranet provides the facility to work in groups and for teleconferences.

# Intranet Advantages

## **Communication**

It provides a cheap and easy communication. An employee of the organization can communicate with another employee through email, chat.

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## **Time-saving**

Information on the intranet is shared in real time, so it is time-saving.

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## **Collaboration**

Collaboration is one of the most important advantage of the intranet. The information is distributed among the employees of the organization and can only be accessed by the authorized user.

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## **Platform independency**

It is a neutral architecture as the computer can be connected to another device with different architecture.

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## **Cost effective**

People can see the data and documents by using the browser and distributes the duplicate copies over the intranet. This leads to a reduction in the cost.

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